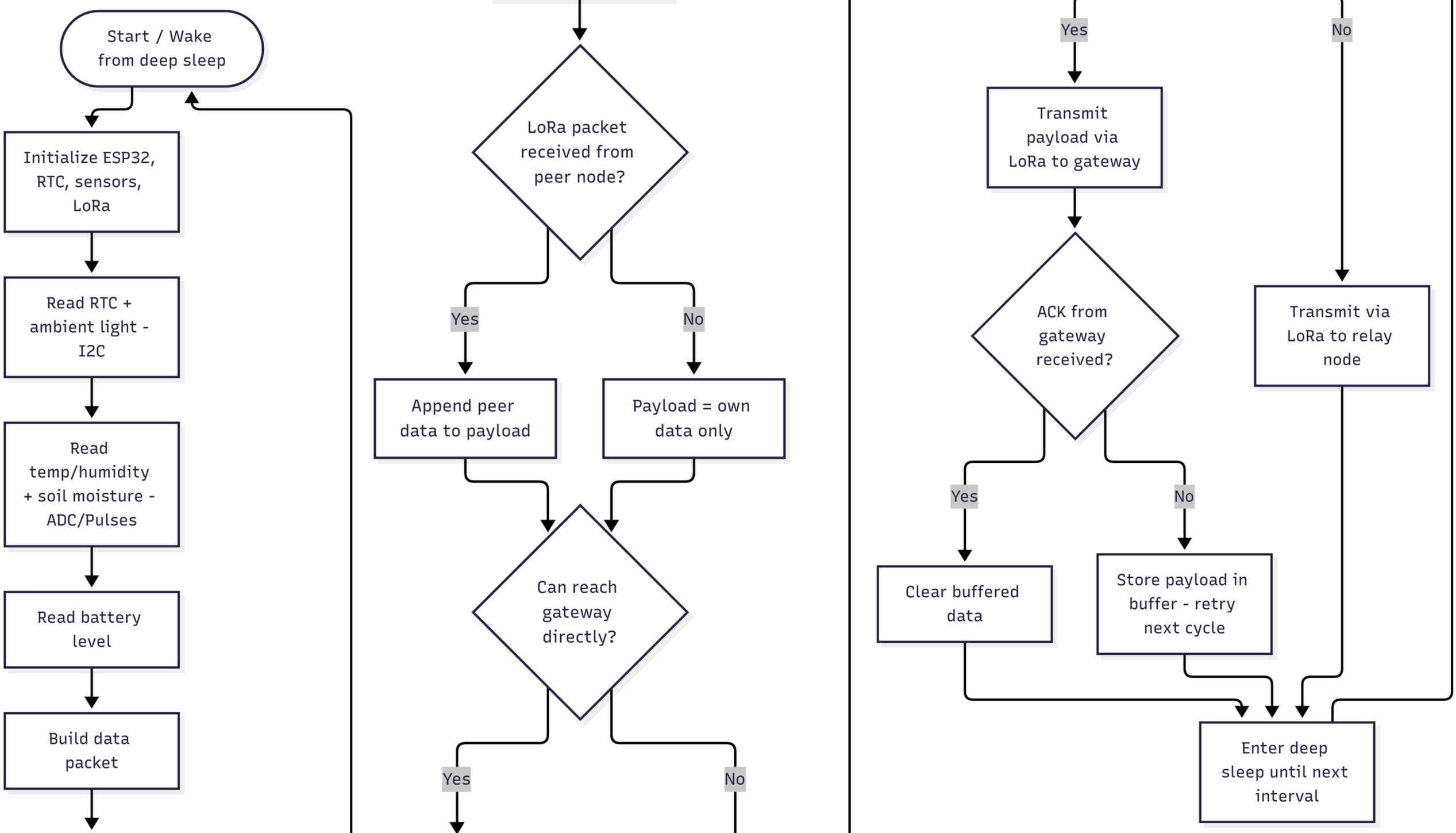

Start / Wake from deep sleep
Initialize ESP32, RTC, sensors, LoRa
Read RTC + ambient light - I2C
Read temp/humidity + soil moisture - ADC/Pulses
Read battery level
Build data packet
LoRa packet received from peer node?
Yes
No
Append peer data to payload
Payload = own data only
Can reach gateway directly?
Yes
No
Yes
No
Transmit payload via LoRa to gateway
ACK from gateway received?
Yes
No
Clear buffered data
Store payload in buffer - retry next cycle
Transmit via LoRa to relay node
Enter deep sleep until next interval